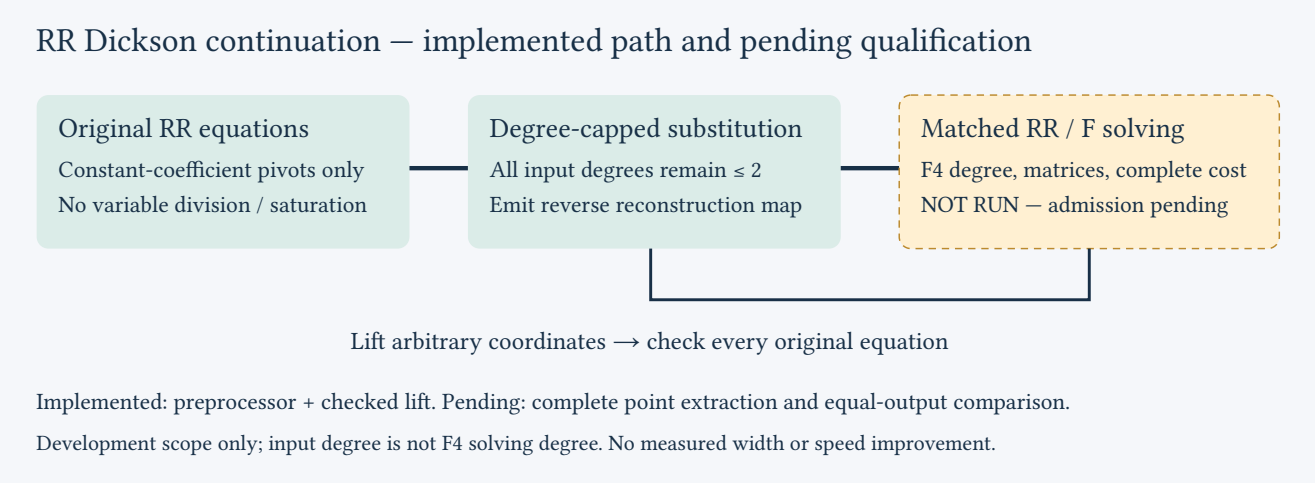
RR Dickson matrix-reduction continuation

10 November 2026 · Implementation report · Producer-unverified

The opt-in native candidate eliminates constant-coefficient pivots only when every remaining input equation stays within the original degree bound. It emits a reverse reconstruction map with original/reduced input hashes. A production lift command reconstructs arbitrary reduced solutions and checks every original equation. No F4 experiment or cost comparison has run.



Why this candidate

The November 7–8 P1 follow-ups report lower RR solving degree with Dickson membership at m = 3, but arm F has narrower matrices. The m = 5 tower cell remains censored. Those observations motivate testing redundant-variable elimination; they do not establish that this candidate helps. msolve may already perform equivalent preprocessing.

Implemented and checked

Two explicit modes: affine pivots only, or constant pivots from equations up to degree two. Neither divides by a variable. Every accepted substitution is checked against all remaining equations for input-degree growth. Native unit tests cover exhaustive small-field solution/lift equivalence, degree-cap and nonconstant-pivot rejection, inconsistent constants, inverse scaling, dependent reconstruction, malformed maps and an existing tiny Dickson fixture. Historical emitter calibration remains in the upstream selftest. These are development checks, not signed PCC verification or a performance measurement.

Missing measurement and qualification

Input degree does not bound F4 solving degree. F has an x-only target; RR fixes sign and uses a monic-function chart. Complete point extraction and chart normalization must precede an equal-output claim. The protocol compares RR, RR-affine, RR-quadratic, F and T on identical inputs, charging transformation, solve, extraction, reconstruction, verification and failed queries.

The checkpoint gate failed after the task resume was filled: no AWS profile or checkpoint upload token is configured. GitHub authentication exists, but that submission route does not waive custody or live budget/pool admission. Budget headroom is unknown. No paid run was dispatched and no local benchmark substituted for it. Development and review continued through this blocker.

Sources and exact next action

Repository sources: research/split_prime_ecdlp_20261107/followups/p1_rr_dickson/ RESULTS.md, PROTOCOL-R2.md, p1tool/README.md; research/ic_tree_gluing_20261103/ round7_rr/README.md and STATUS.md. Primary source: Petit–Kosters–Messeng, PKC 2016. KB review unavailable: no loaded MCP and repository wrapper reports no executable.

Resume from HANDOFF.md. Restore checkpoint transport, verify live budget and pool capacity, qualify the complete-output normalizer, then freeze the input manifest and source-built image before the matched m3 panel. Success requires retained lower solving degree, smaller matrices and lower complete equal-output cost. No such outcome is asserted here. Full details: README.md, PROTOCOL.md, checks/native-tests.txt and the linked problem record. Historical scoreboards remain unchanged because there is no new scientific measurement to plot.